

Cloud Computing (Service) Models

The background of the slide features several thick, wavy, overlapping lines in orange, light blue, and lime green, creating a dynamic and modern aesthetic.

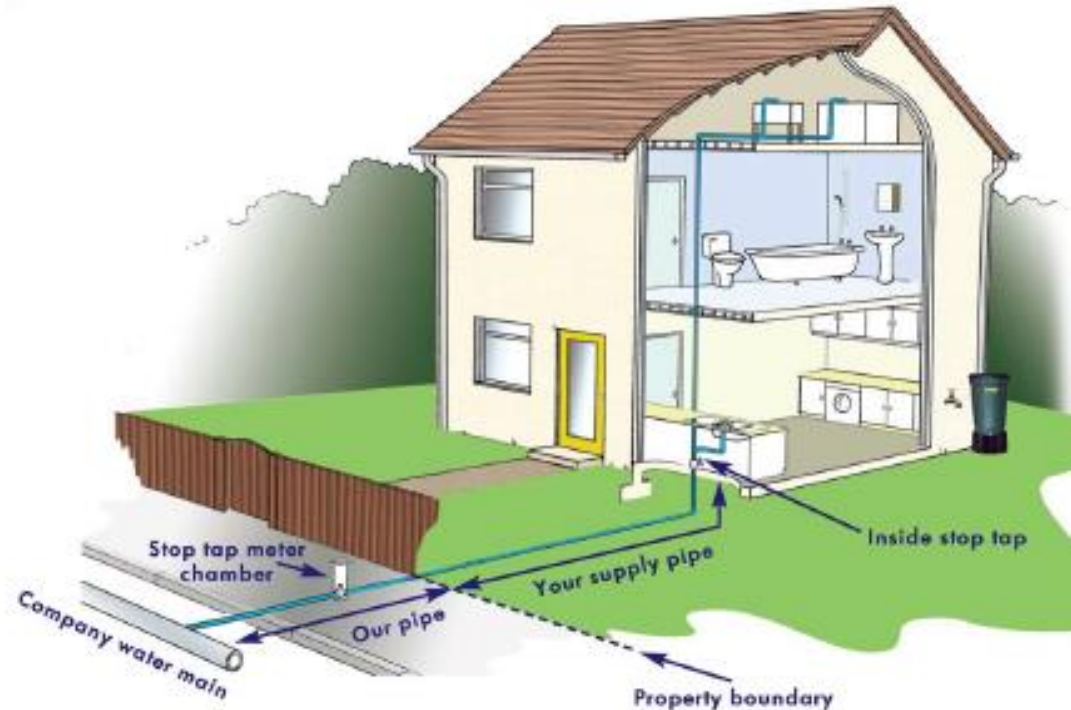
Life Style: Utility Service

- Power/ heat/electricity/water supply to your home

Before



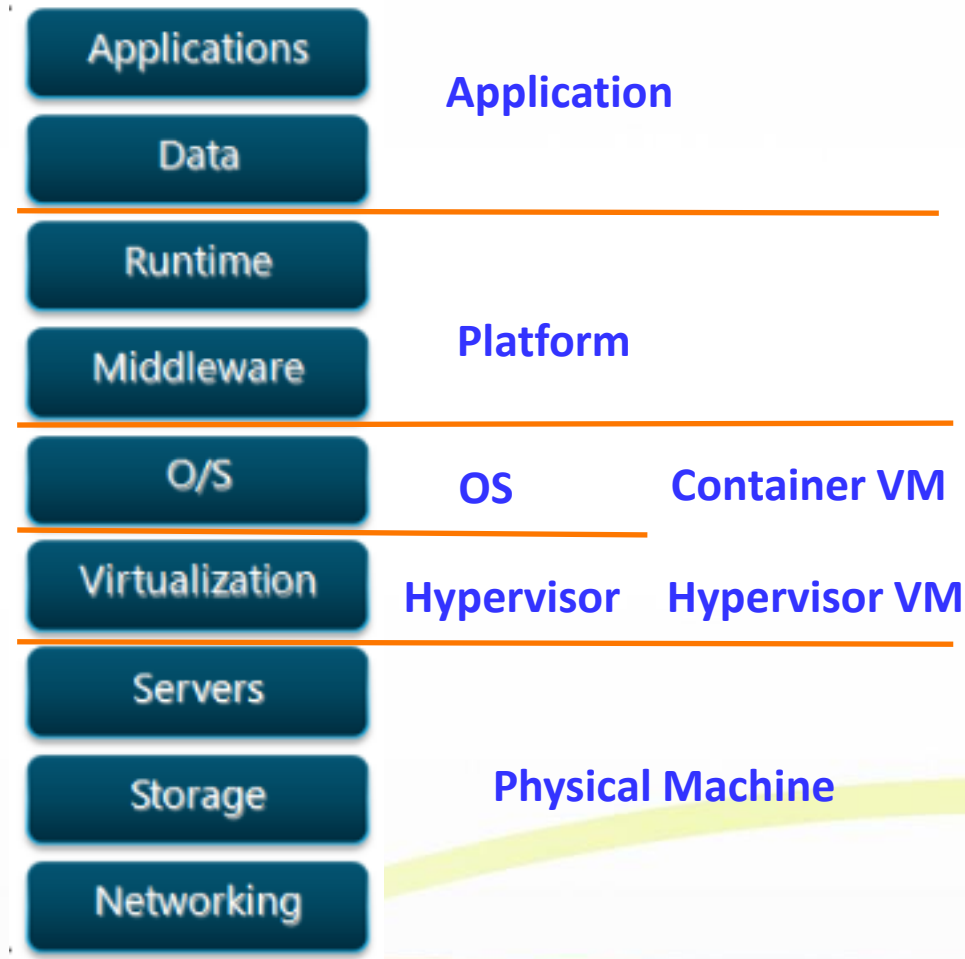
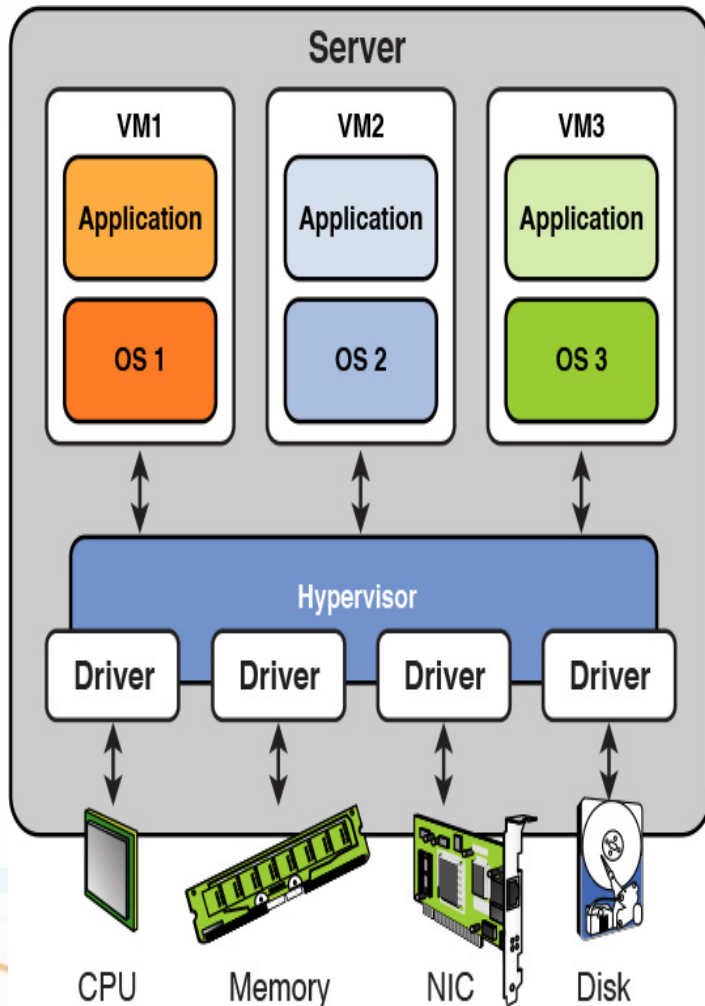
Now



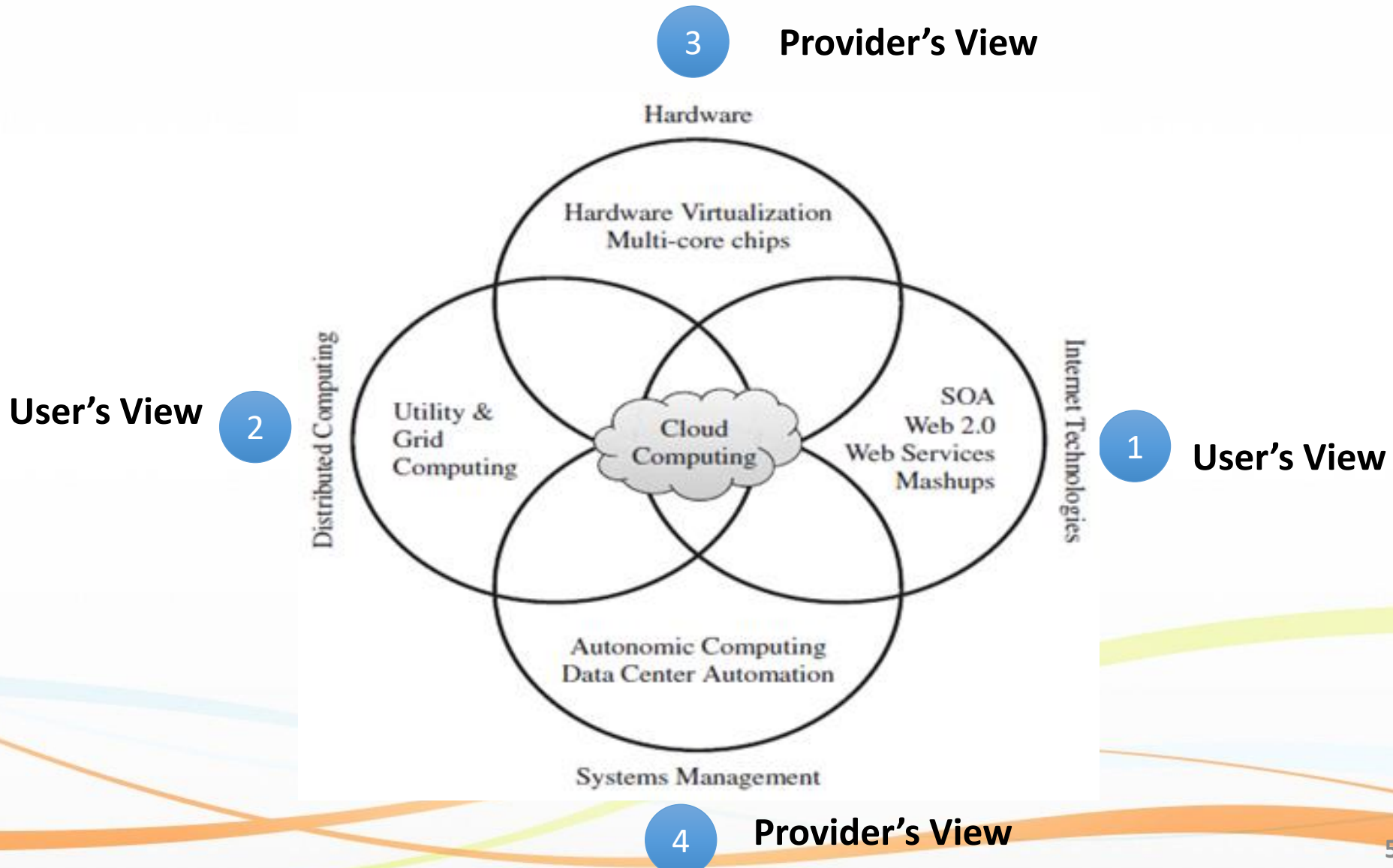
Life Style: Transportation Service



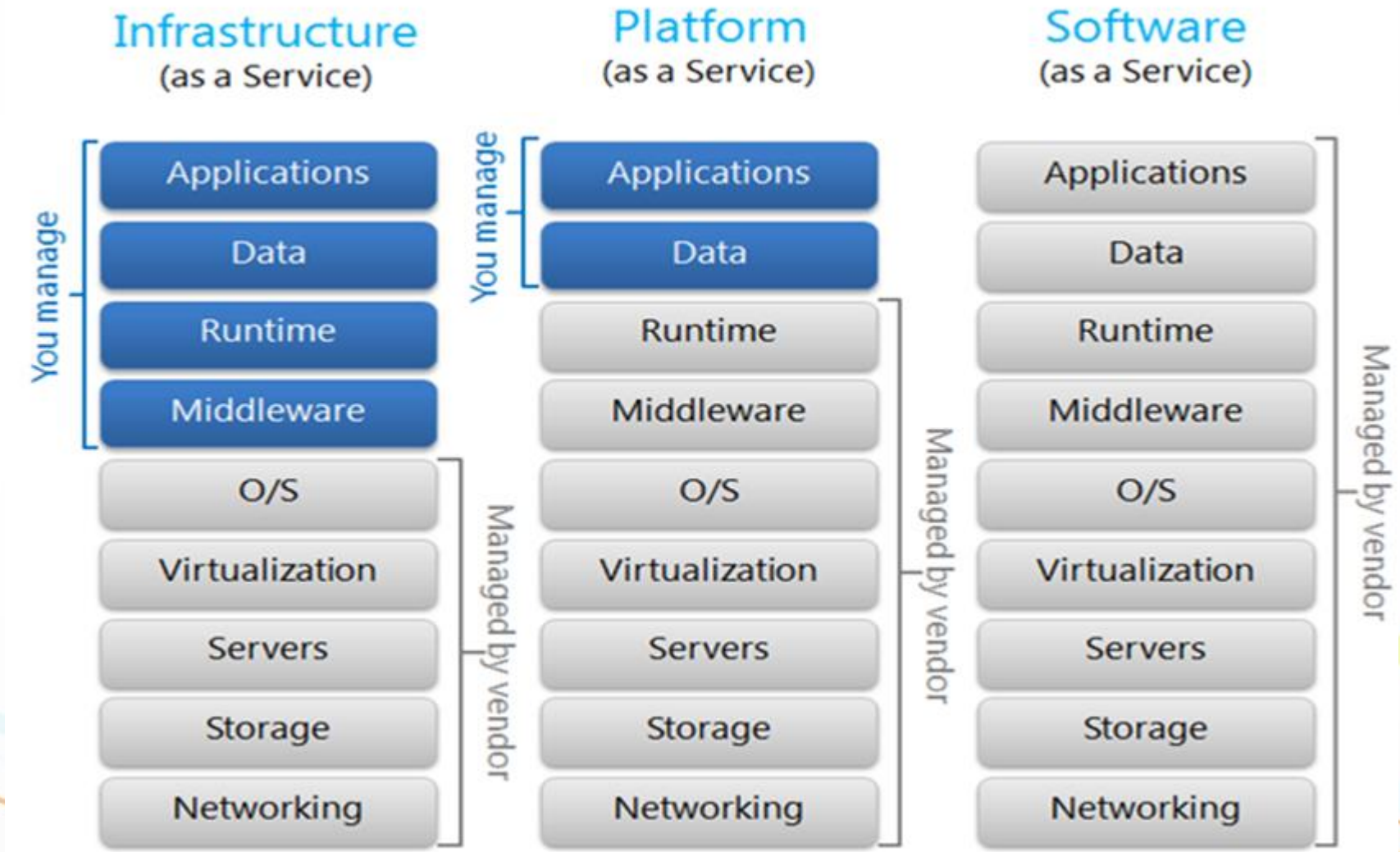
Computing Service Level



Roots of Cloud Computing



Cloud Computing Service Models

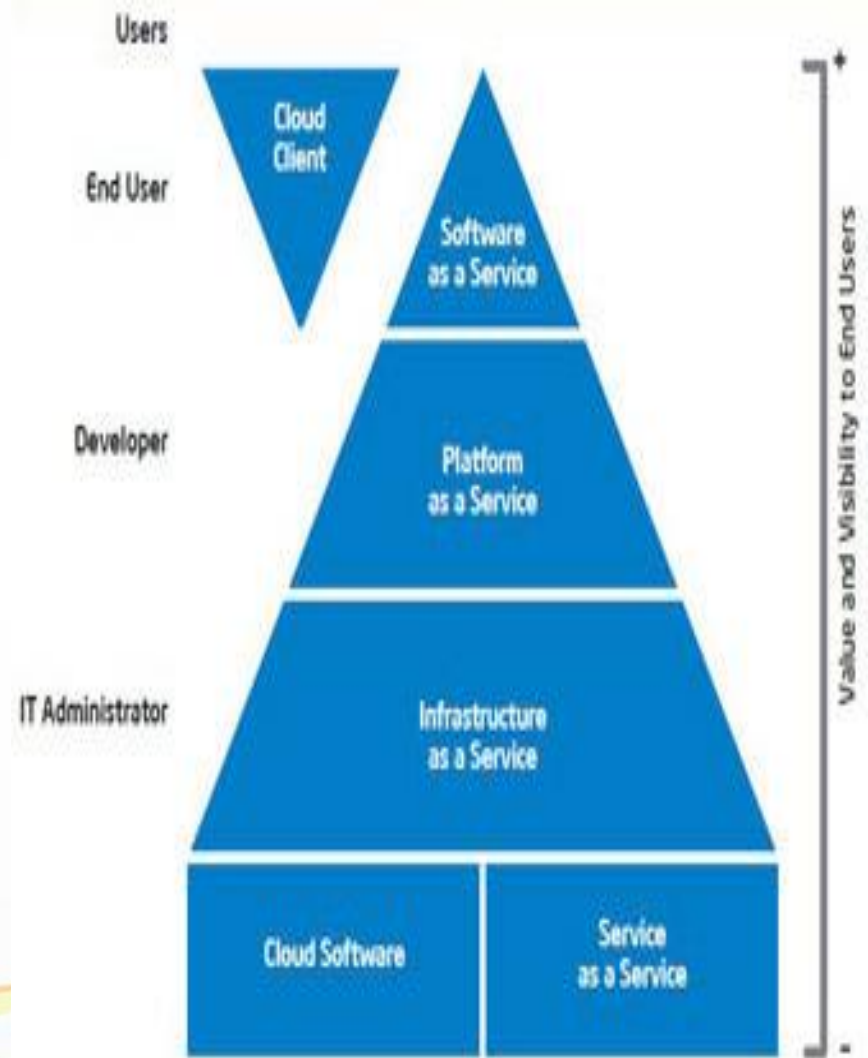


Layers and Types of Clouds

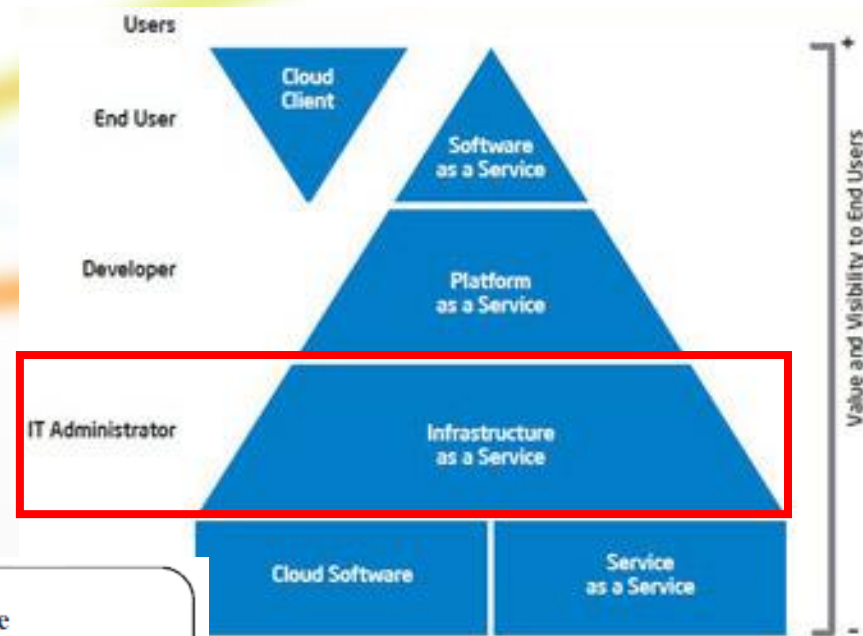
Service Class	Main Access & Management Tool	Service content
 SaaS	Web Browser	Cloud Applications Social networks, Office suites, CRM, Video processing
 PaaS	Cloud Development Environment	Cloud Platform Programming languages, Frameworks, Mashups editors, Structured data
 IaaS	Virtual Infrastructure Manager	Cloud Infrastructure Compute Servers, Data Storage, Firewall, Load Balancer

FIGURE 1.3. The cloud computing stack.

Source: <http://itcenter.intel.com/ResourceLibrary>



Infrastructure as a Service (IaaS)



IaaS

Virtual
Infrastructure
Manager

Cloud Infrastructure

Compute Servers, Data Storage,
Firewall, Load Balancer

17

Source: <http://itcenter.intel.com/ResourceLibrary>

IaaS

▶ Providing infra resource service to users

❖ Virtual Servers

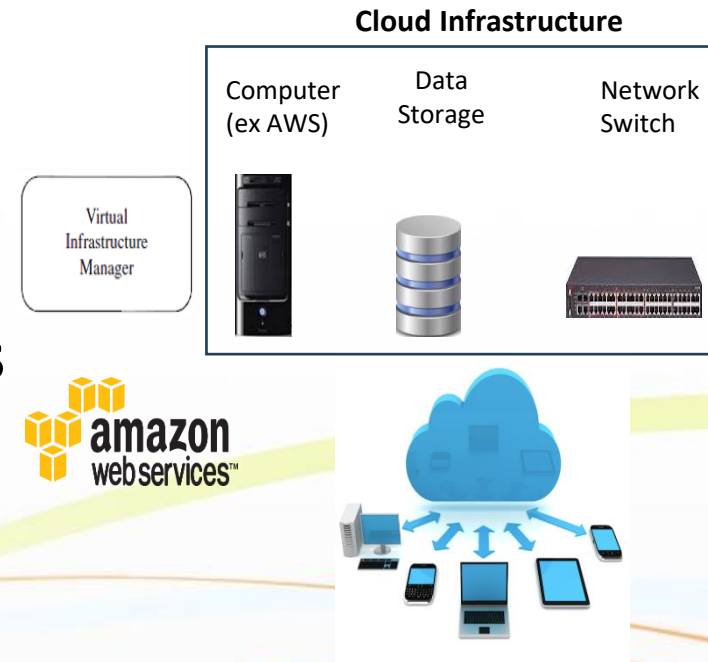
- Containing one or more CPUs and GPU
- Running several choices of
 - Operating systems
 - Linux, Windows, etc.

❖ Virtual Storage Space

- HDD, SSD, Memory, Database, etc.

❖ Virtual Communication Facilities

- IP, VPN , L4 Switch etc.



Infrastructure as a Service Providers

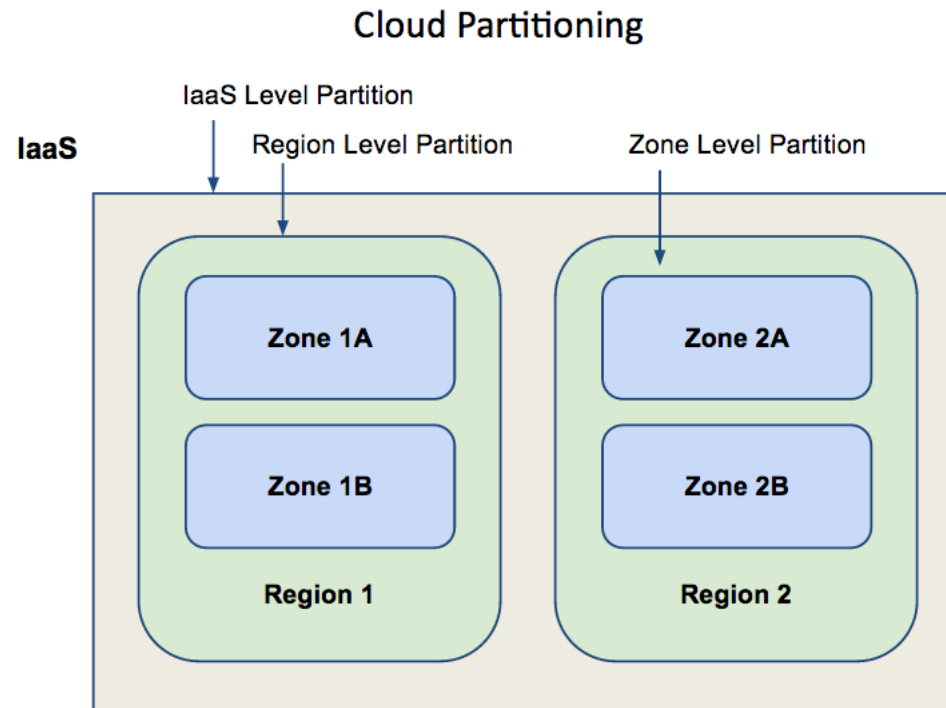
- ▶ Offering **virtualized resources** (computation, storage, and communication) on demand
- ▶ Features
 - ❖ Geographic Presence
 - ❖ User Interfaces and Access to Servers
 - ❖ Advance **Reservation** of Capacity
 - ❖ **Automatic Scaling and Load Balancing**
 - ❖ **Service-Level Agreement**
 - ❖ **Hypervisor and Operating System Choice**
 - ❖ **Image and Configuration Management**
 - ❖ **Physical Resource(Storage, Network, etc.) Management**

IaaS Features

► Geographic Presence

❖ **Resource** will be **provisioned** through **Distributed Several Data Centers** to improve **availability** and **responsiveness**

➤ For example, *Amazon Web Services* presents the concept of “*availability zones*” and “*regions*” for its EC2 service



IaaS

▶ Provides multiple access approaches

❖ Graphical User Interfaces (GUI)

- **Launch, customize, and monitor a few virtual servers**
- Do not necessary need to repeat the process several times

❖ Command-Line Tools (CLI)

- Start and shutdown a number of virtual servers at regular intervals
- More flexibility and the possibility of **automating repetitive tasks via scripts**

❖ Web Service (WS) APIs

- Programmatic access to a cloud using standard HTTP requests
- **Allowing complex services to be built on top of IaaS clouds**

IaaS Features

▶ Advance Reservation of Capacity

❖ Reserve resources for a specific time frame in the future

➤ For example, **Amazon Reserved Instances(ARI)**

- Allowing users to **pay a fixed amount** of money in advance to **guarantee resource availability at anytime during an agreed period** and then paying a discounted hourly rate when resources are in use
- **However, only long periods of 1 to 3 years** are offered; therefore, users cannot express their reservations in finer granularities—for example, hours or days.

➤ However, **most clouds only support best-effort requests**; that is, users requests are served whenever resources are available

IaaS Features

▶ Automatic Scaling and Load Balancing

❖ Automatic Elasticity

- **Automatic scaling up and down, based on application-specific metrics**
- When the number of virtual servers is increased by automatic scaling, incoming **traffic** must be **automatically distributed among the available servers**

IaaS Features

- ▶ Service-Level Agreements (SLAs)
 - ❖ Delivery of a certain **QoS**
 - ❖ **Availability** and **performance guarantees**
 - ❖ **Most IaaS providers focus** their SLA terms **on availability guarantees**, specifying the minimum percentage of time the system will be available during a certain period
 - For instance, Amazon EC2 states that “if the annual uptime Percentage for a customer drops below **99.95%** for the service year, that customer is eligible to receive a service credit equal to 10% of their bill.”

IaaS Features

▶ Hypervisor and Operating System Choice

❖ Hypervisors

➤ Type 1: **Native or Bare-Metal Hypervisors**

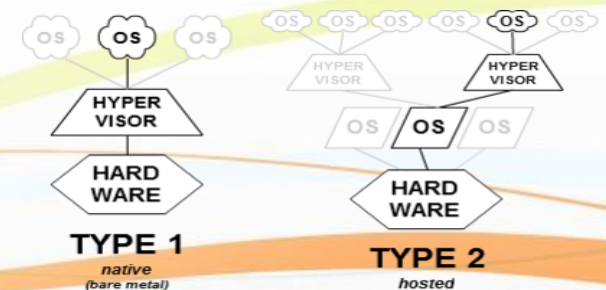
- Hypervisors run directly on the host's hardware to control the hardware and to manage guest operating systems
 - **Xen**, Oracle VM Server for SPARC, Oracle VM Server for x86, the Citrix XenServer, Microsoft Hyper-V and VMware ESX/ESXi, etc.

➤ Type 2: **Hosted Hypervisors (Kernel Mode, User Mode)**

- Hypervisors run on a conventional operating system just as other computer programs do
 - **KVM(Kernel Mode)**, VMware Workstation, VMware Player, **VirtualBox(User Mode)**, Parallels Desktop for Mac and QEMU, etc.

❖ Operation Systems

- Linux, Windows, etc.



Case Studies: Amazon Web Services

- Commercial Solution -

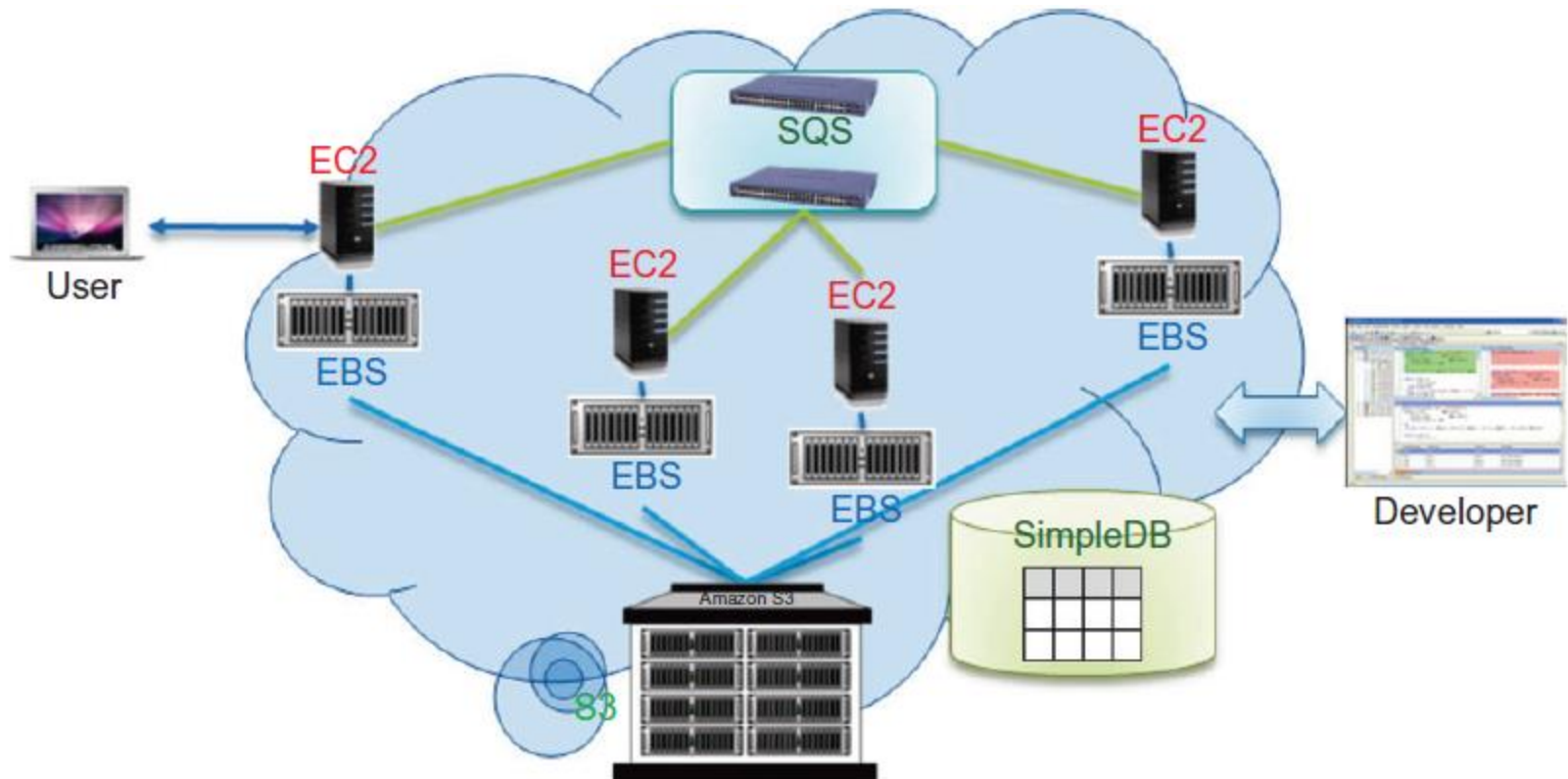
- ▶ One of the **major players in the cloud computing market**
- ▶ Pioneered the introduction of IaaS clouds in **2006**
- ▶ Main Services Offered by AWS

Service Area	Service Modules and Abbreviated Names
Compute	Elastic Compute Cloud (EC2), Elastic MapReduce, Auto Scaling
Messaging	Simple Queue Service (SQS), Simple Notification Service (SNS)
Storage	Simple Storage Service (S3), Elastic Block Storage (EBS), AWS Import/Export
Content Delivery	Amazon CloudFront
Monitoring	Amazon CloudWatch
Support	AWS Premium Support
Database	Amazon SimpleDB, Relational Database Service (RDS)
Networking	Virtual Private Cloud (VPC) (Example 4.1, Figure 4.6), Elastic Load Balancing
Web Traffic	Alexa Web Information Service, Alexa Web Sites
E-Commerce	Fulfillment Web Service (FWS)
Payments and Billing	Flexible Payments Service (FPS), Amazon DevPay
Workforce	Amazon Mechanical Turk

Case Studies: Amazon Web Services

- Commercial Solution -

► AWS Architecture



Case Studies: Microsoft Azure

- Commercial Solution -

▶ Brief History

- ❖ Announced in October 2008
- ❖ Released on 1 February 2010 as Windows Azure
- ❖ Renamed to Microsoft Azure on 25 March 2014

▶ Main Services

❖ Compute

- Virtual Machines, Web Sites

❖ Data Management

- SQL, Storage Blob, Storage Table, File Service

❖ Networking

- Virtual Network, Traffic Manager, Express Route

❖ Developer & IT Service

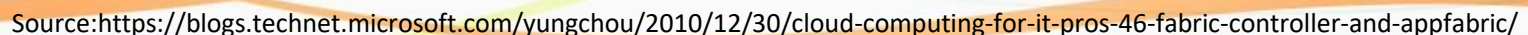
- Virtual Studio Online, Azure SDK

❖ Big Compute & Big Data

- HDInsight(Hadoop), High Performance Computing (HPC)

- Commercial Solution -

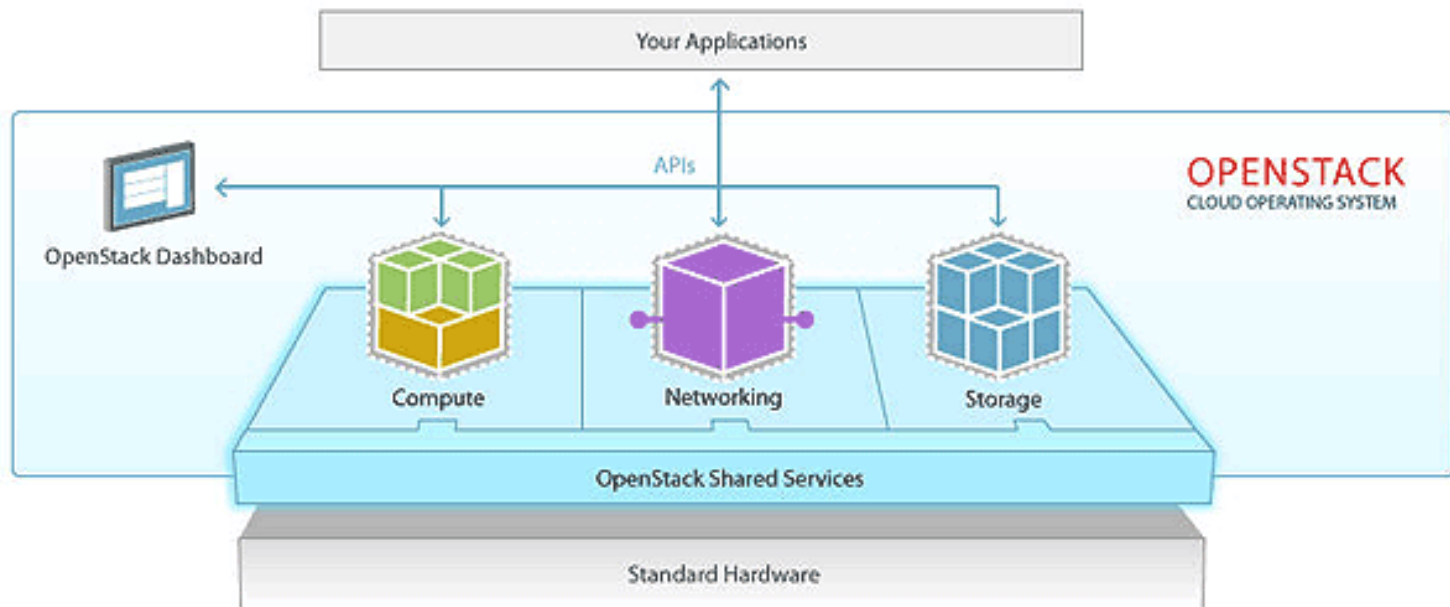
Windows Azure Compute Node



Case Studies: Open Stack

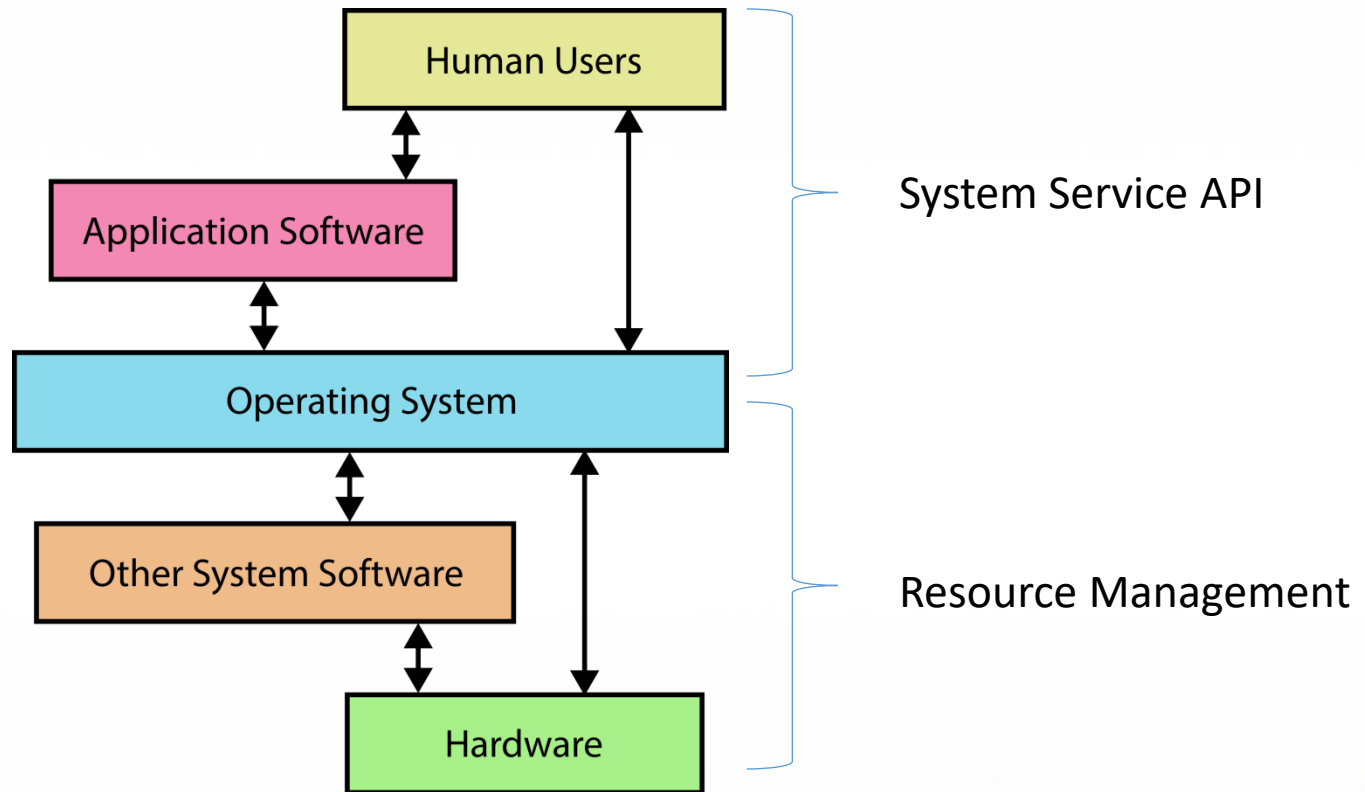
- Open Source Solution -

- ▶ Open source cloud management software platform for IaaS
- ▶ Began in 2010 as a joint project of Rackspace Hosting and NASA and then managed by OpenStack Foundation
- ▶ More than 500 companies have joined the project



O.S

► O.S.

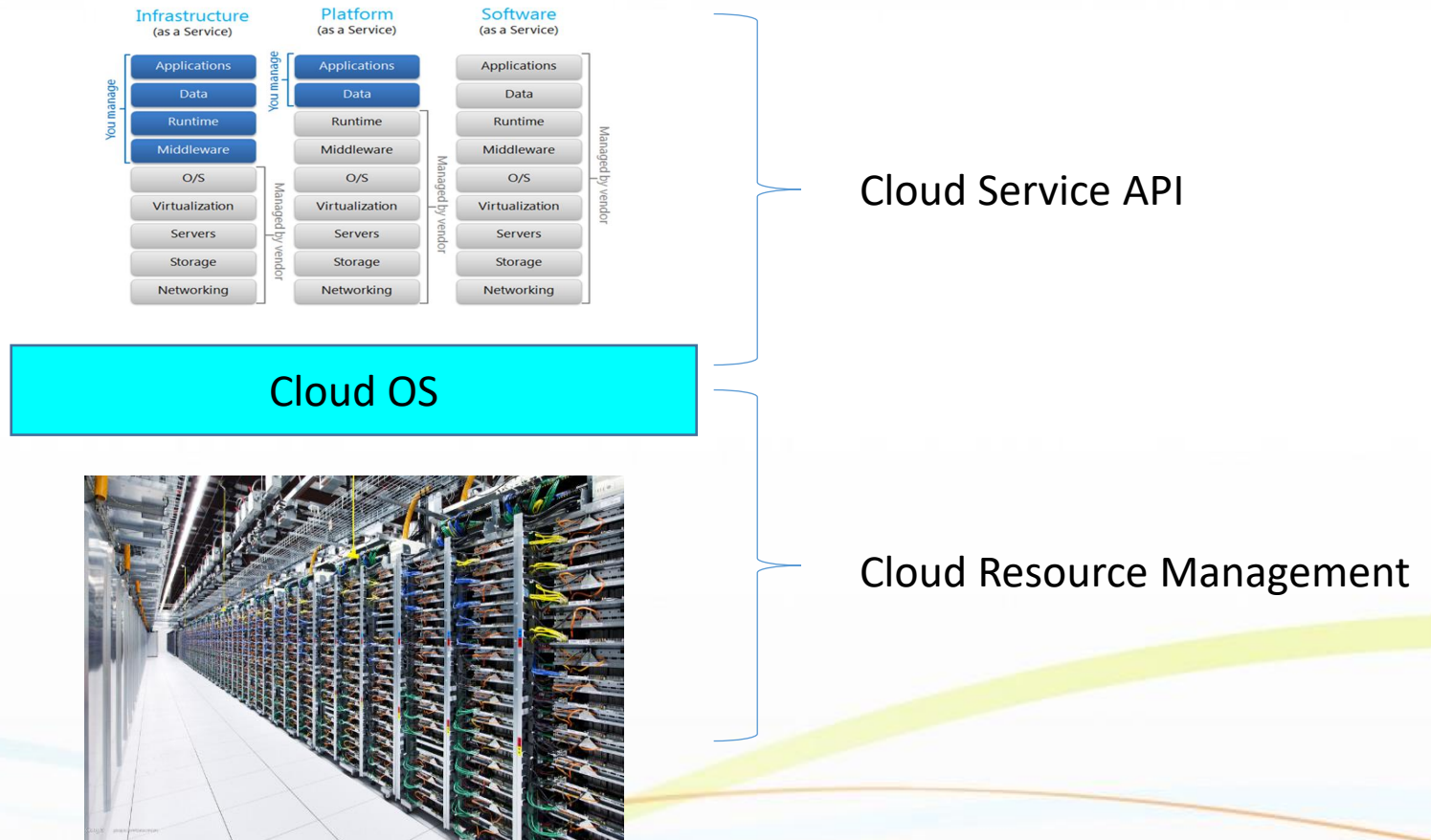


Operating System (OS) : a set of programs that

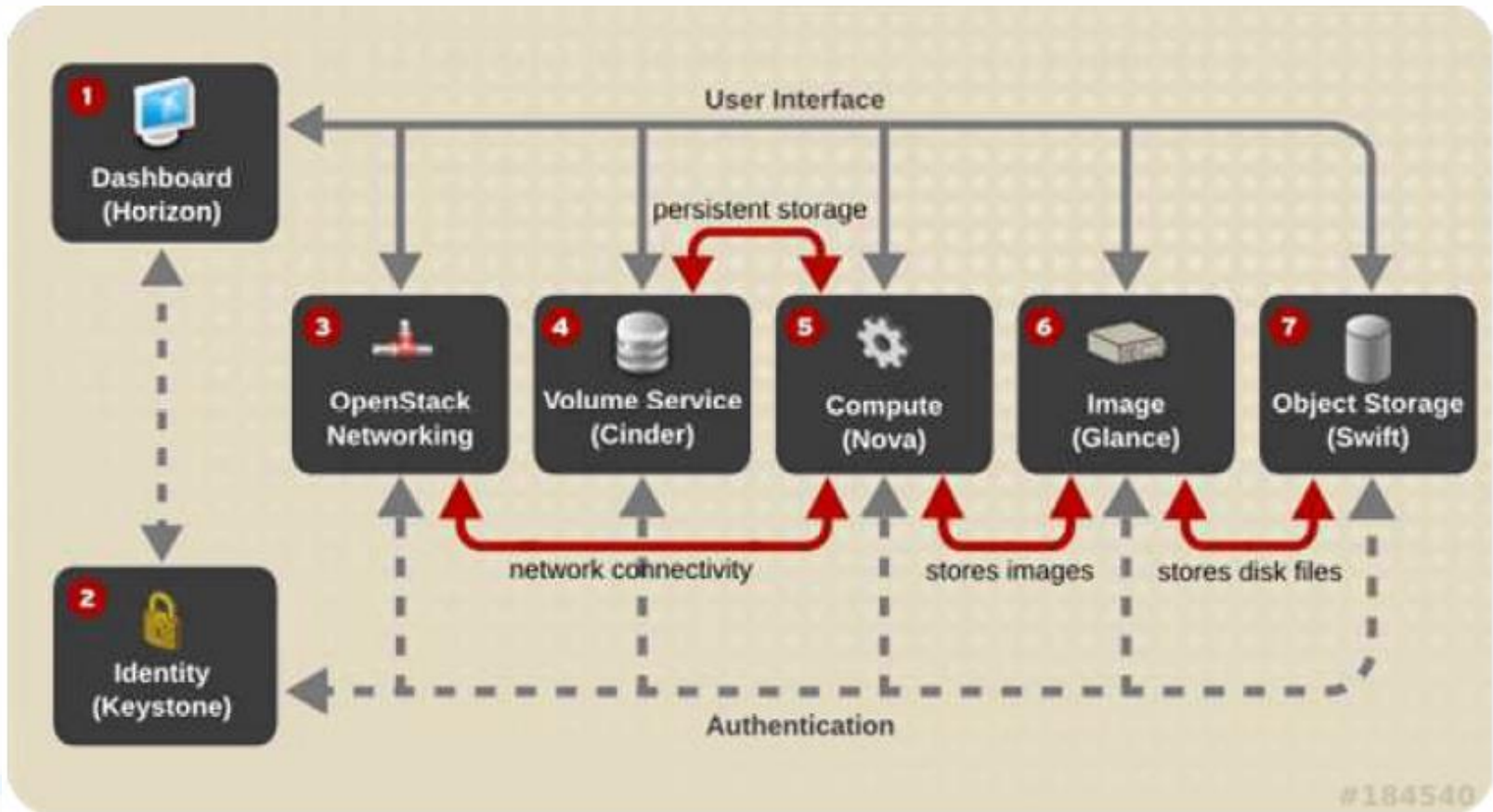
- manage computer hardware resources and
- provide common services for application software.

Cloud O.S.

► Cloud O.S.



OpenStack Architecture

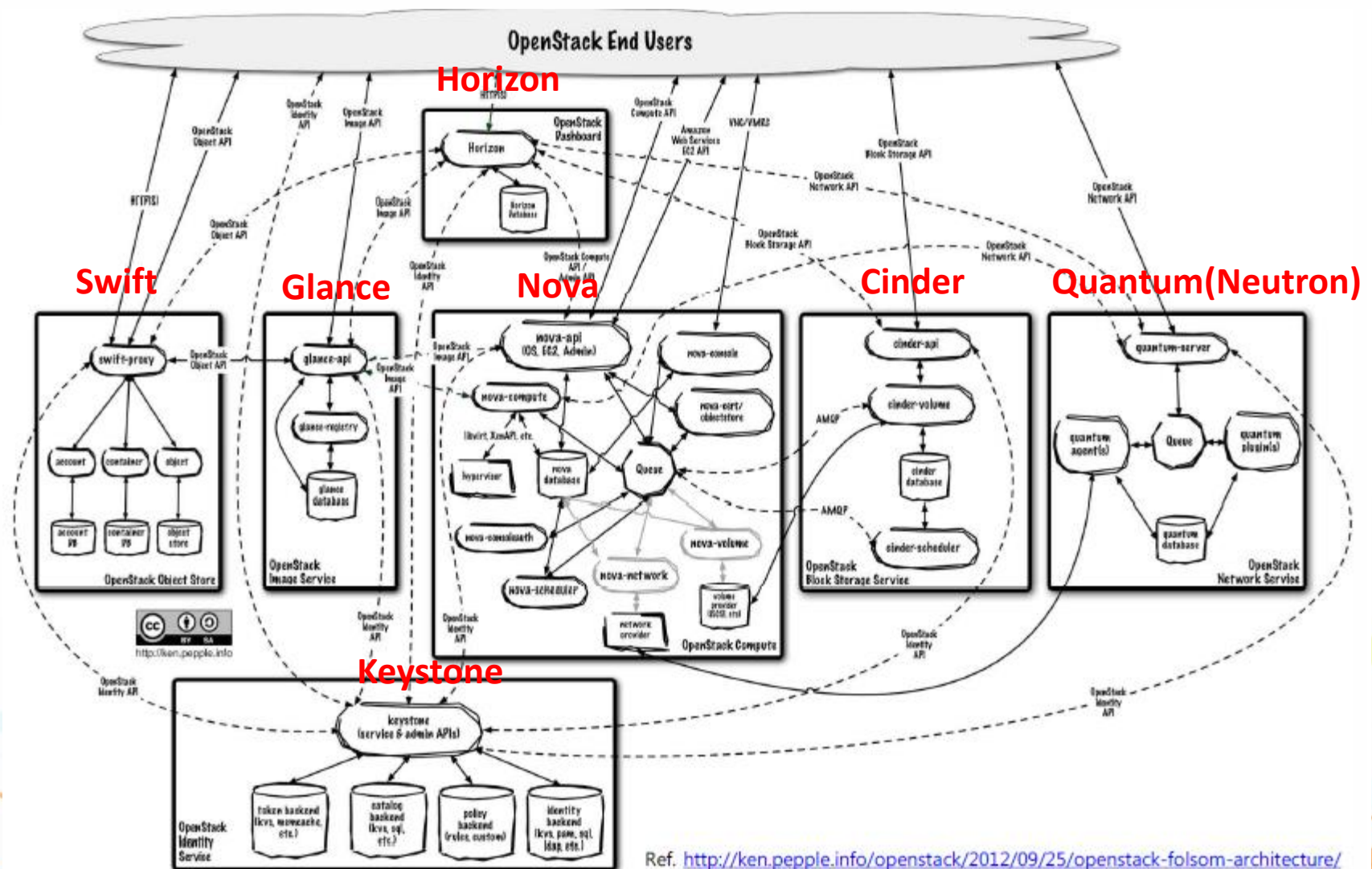


Ref. https://access.redhat.com/documentation/en-US/Red_Hat_Enterprise_Linux_OpenStack_Platform/2/html/Getting_Started_Guide/ch01.html

OpenStack Components

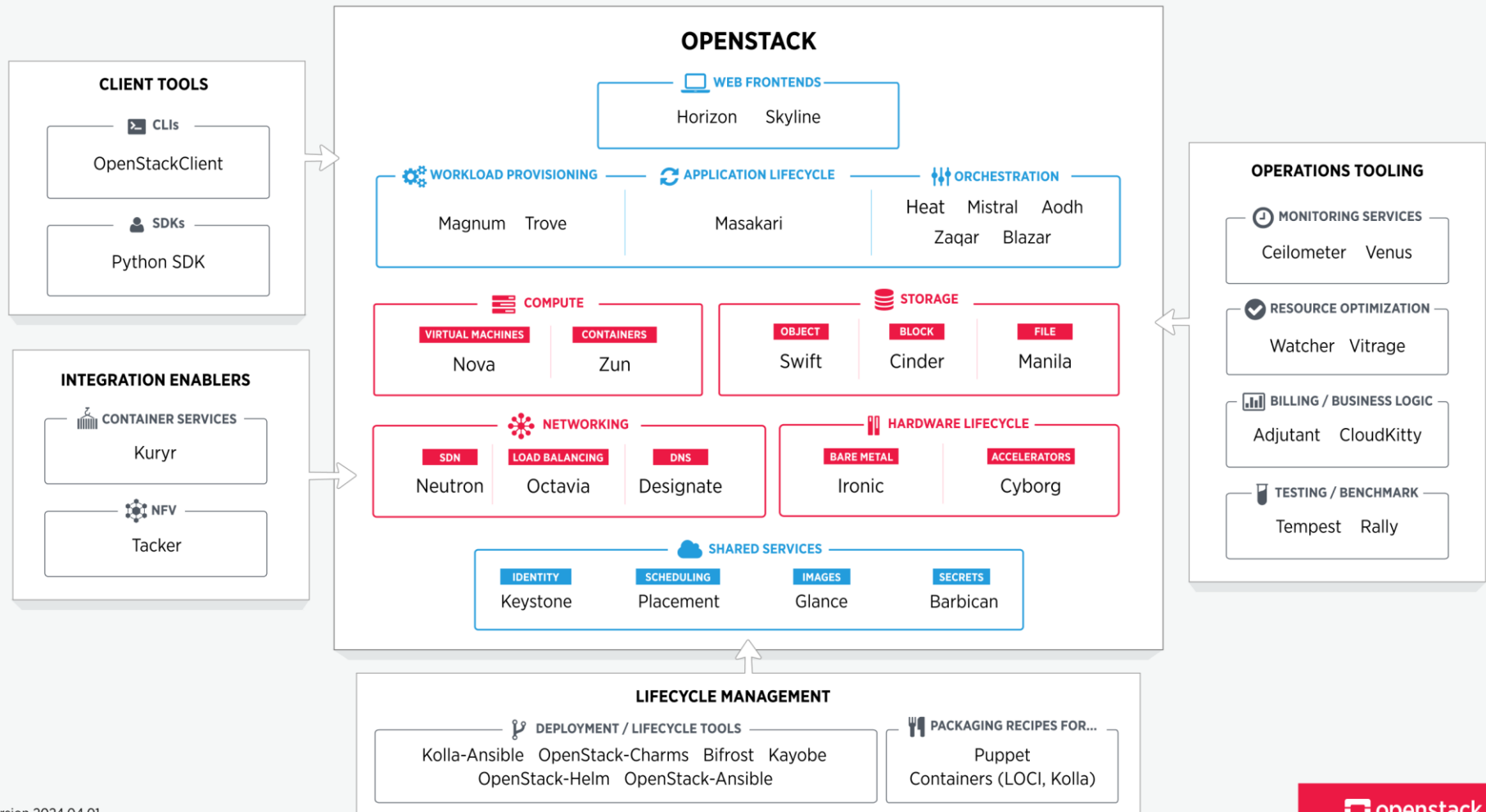
Services	Project Name	Description
Dashboard	Horizon	Web Based self service
Compute	Nova	Manages the lifecycle of compute instances
Networking	Neutron	Provides network connectivity as service for openstack service
Storage		
Object storage	Swift	Stores and retrieves unstructured data objects via RESTful API, HTTP based API
Block storage	Cinder	Provides persistent block storage to running instances
Shared Services		
Identity Services	Keystone	Provides authentication and authorization for open stack services
Image Service	Glance	Stores and retrieves VM disk images
Telemetry	Celometer	Monitors and metering service for billing, scalability, benchmarking and statistical purpose.
Higher-level Service		
Orchestration	Heat	Orchestration of multiple composite cloud applications
Database service	Trove	Database service for relational and non-relational database engines

OpenStack Architecture (Detained)



Open Stack Landscape

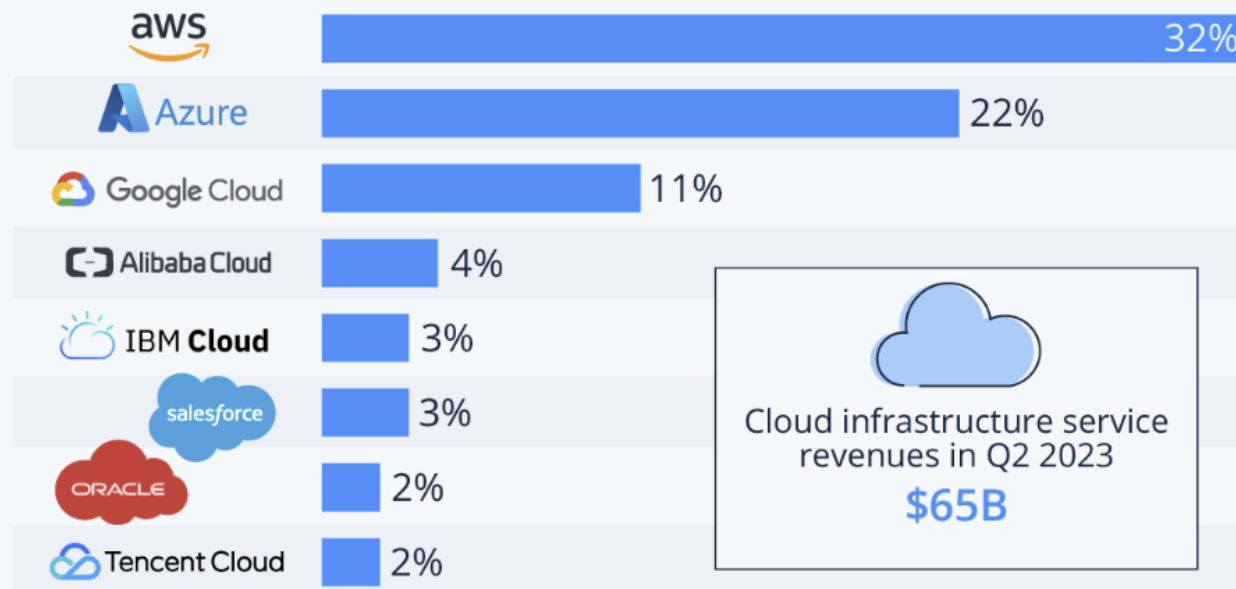
<https://www.openstack.org/software/>



Market Share of IaaS

Amazon Maintains Lead in the Cloud Market

Worldwide market share of leading cloud infrastructure service providers in Q2 2023*



* Includes platform as a service (PaaS) and infrastructure as a service (IaaS)

Platform as a Service (PaaS)



Windows Azure™

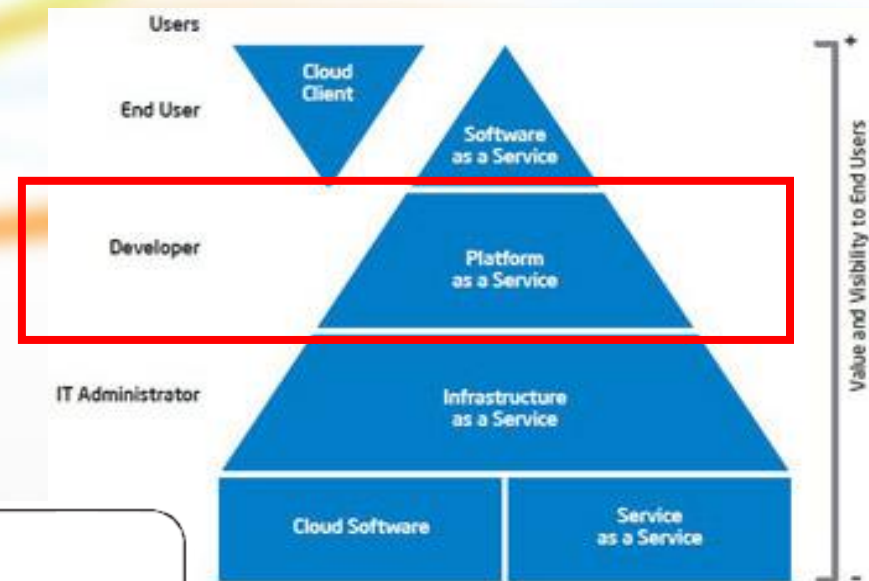


PaaS

Cloud
Development
Environment

Cloud Platform

Programming languages, Frameworks,
Mashups editors, Structured data



Source: <http://itcenter.intel.com/ResourceLibrary>

PaaS Model

- ▶ PaaS commonly offers a **development and deployment environment**
 - ❖ Allow users to create and run their applications with **little or no concern to low-level details of the platform**
 - ❖ Support **specific programming models, languages and frameworks**
 - ❖ **Persistent data storage and in-memory caches**

SW Development
Platform

Media Production
Platform

Data Analytic
Platform

BigData Processing
Platform

AI Processing
Platform

Robot Development
Platform

Android/IOS App
Development Platform

etc

Cloud Platforms

PaaS Features

- ▶ Programming models aim at efficiently solving a particular problem
 - ❖ Processing of large dataset in clusters of computers (MapReduce Model(Data Parallel Computing Model))
 - ❖ Development of request-based Web services and applications(Web Service Model)
 - ❖ Definition and orchestration of business processes in the form of workflows (Workflow Model)
 - ❖ High-performance distributed execution of various computational tasks(Traditional HPC Model)
 - ❖ AI Processing Model (Matrix Parallel Computing Model, Distributed Parallel Computing Model)

PaaS Features

▶ Multiple Programming Languages

- ❖ Python and Java (e.g., Google AppEngine)
- ❖ .NET languages (e.g., Microsoft Azure)
- ❖ Ruby (e.g., Heroku)

▶ Software Frameworks

- ❖ Web and enterprise application
 - Ruby on Rails, Spring, Java EE, and .NET
- ❖ AI applications
 - Pytorch, Tensorflow, Keras etc

▶ Persistence Options

- ❖ A persistence layer is essential to allow applications to record their state and recover it in case of crashes, as well as to store user data
- ❖ Traditionally, Web and enterprise application developers have chosen relational databases as the preferred persistence method.

PaaS Market Share

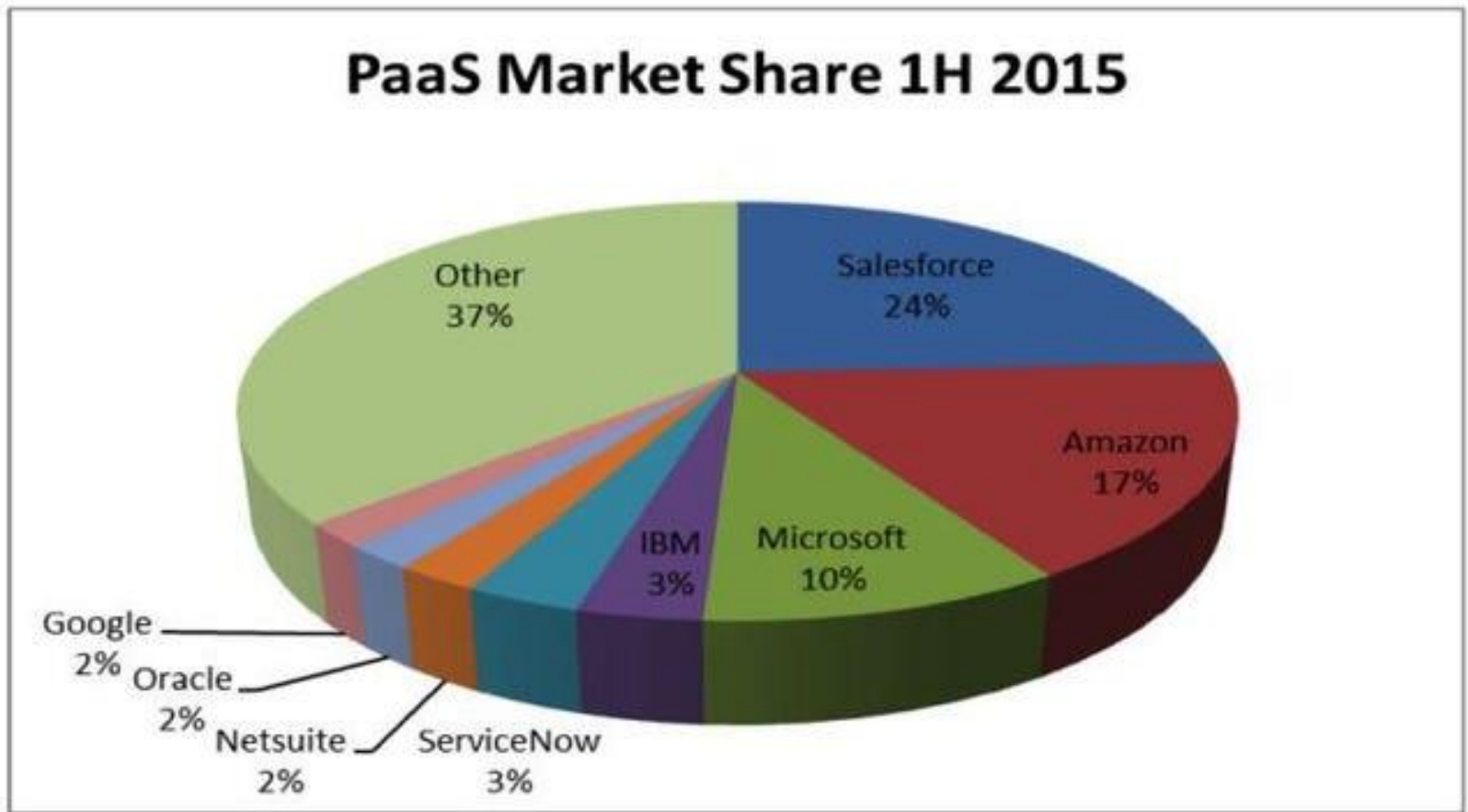
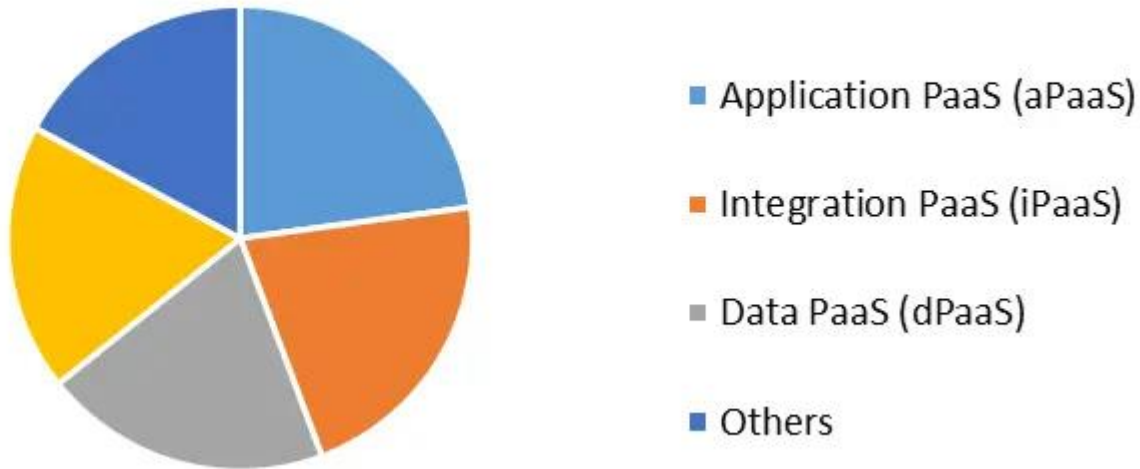


Figure 2: PaaS Market Share, 1H 2015
Source: Wikibon 2015

PaaS Market Share

Global Platform as a Service (PaaS) Market Share, By Type in 2022 (%)



Software as a Service (SaaS)

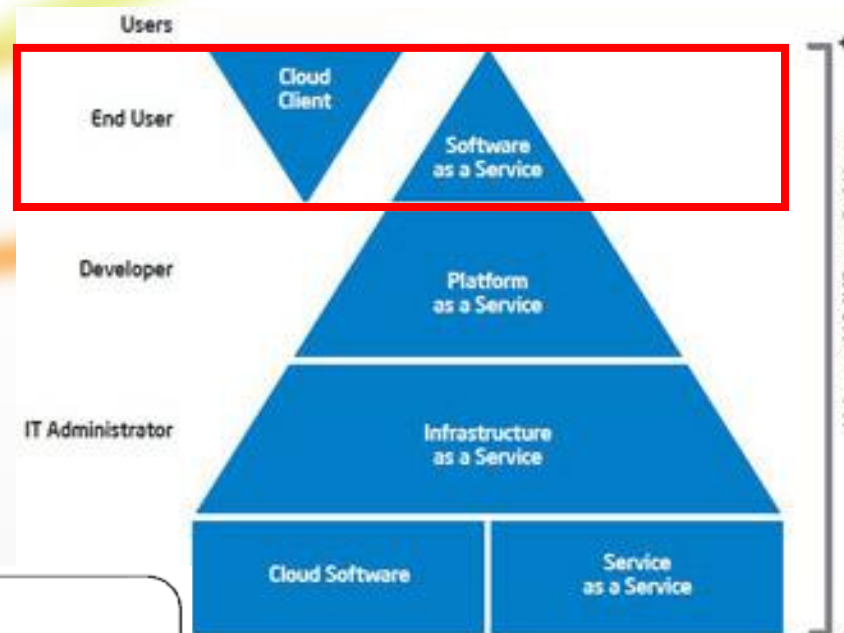


SaaS

Web Browser

Cloud Applications

Social networks, Office suites, CRM,
Video processing



Source: <http://itcenter.intel.com/ResourceLibrary>

SaaS Model

- ▶ The SaaS model **provides software applications** as a service
 - ❖ On the customer side, there is **no upfront investment in servers or software licensing**
 - ❖ On the provider side, costs are kept rather low, compared with conventional hosting of user applications
- ▶ Refers **to browser-initiated application** software over thousands of cloud customers.

End User can access to “Cloud Application” in Service Level

SaaS Features

► Configuration and Customization

❖ **application customization**

- A single customer can alter the set of configuration options (a.k.a. parameters) that affect **its functionality and look-and-feel**
- For example: application appears to be having the **customer's brand**

► Open **integration protocols**

- ❖ Offering integration protocols and application programming interfaces (APIs) that operate over a wide area network
- Typically, these are protocols based on HTTP, REST and SOAP

SaaS Features

- ▶ Collaborative (and "social") functionality
 - ❖ SaaS applications offer features that let its users collaborate and share information
 - ❖ For example
 - Many **project management applications** delivered in the SaaS model offer collaboration features letting users comment on tasks and plans and share documents within and outside an organization
- ▶ Accelerated Feature Delivery
 - ❖ **Application update is decided and executed by the provider, not by customers**
 - ❖ The application only has a single configuration, making development testing faster

Examples SaaS

- ▶ Google Gmail and docs
- ▶ Microsoft SharePoint
- ▶ Salesforce.com(CRM Software)
- ▶ Eli Lilly Company
 - ❖ DNA sequence analysis for discovering new drugs
 - ❖ Using Amazon's AWS platform with provisioned server and storage clusters to conduct high-performance biological sequence analysis without using an expensive supercomputer
- ▶ The New York Times
 - ❖ Using Amazon's EC2 and S3 services to retrieve useful pictorial information quickly from millions of archival articles and newspapers

SaaS Market

Top enterprise SaaS vendors by percent annual growth



Adoption rate of applications fully run in public cloud

Source: Spiceworks



Market Share of IaaS, PaaS, SaaS

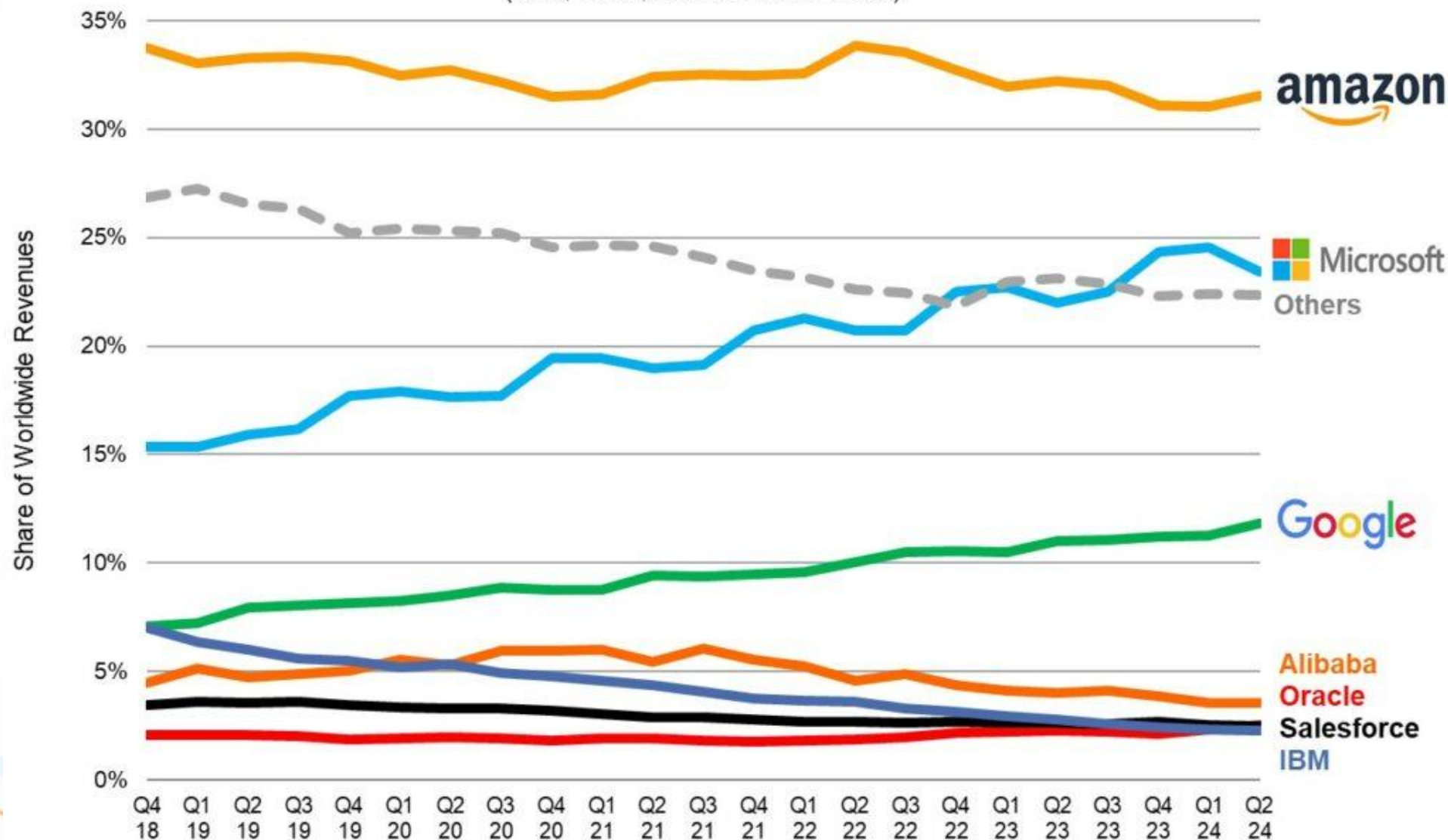
Worldwide Public Cloud Services Revenue and Year-over-Year Growth, Calendar Year 2023 (revenues in US\$ billions)

Deployment Category	2023 Revenue	2023 Market Share	2022 Revenue	2022 Market Share	2023/2022 Growth
IaaS	\$133.4	19.9%	\$115.5	20.7%	15.6%
PaaS	\$123.3	18.4%	\$95.4	17.1%	29.3%
SaaS – Applications	\$298.5	44.6%	\$254.4	45.6%	17.4%
SaaS – System Infrastructure Software	\$114.0	17.0%	\$93.1	16.7%	22.5%
Total	\$669.2	100%	\$558.3	100%	19.9%

Source: IDC Worldwide Semiannual Public Cloud Services Tracker, 2H 2023, May 2024.

Cloud Provider Market Share Trend

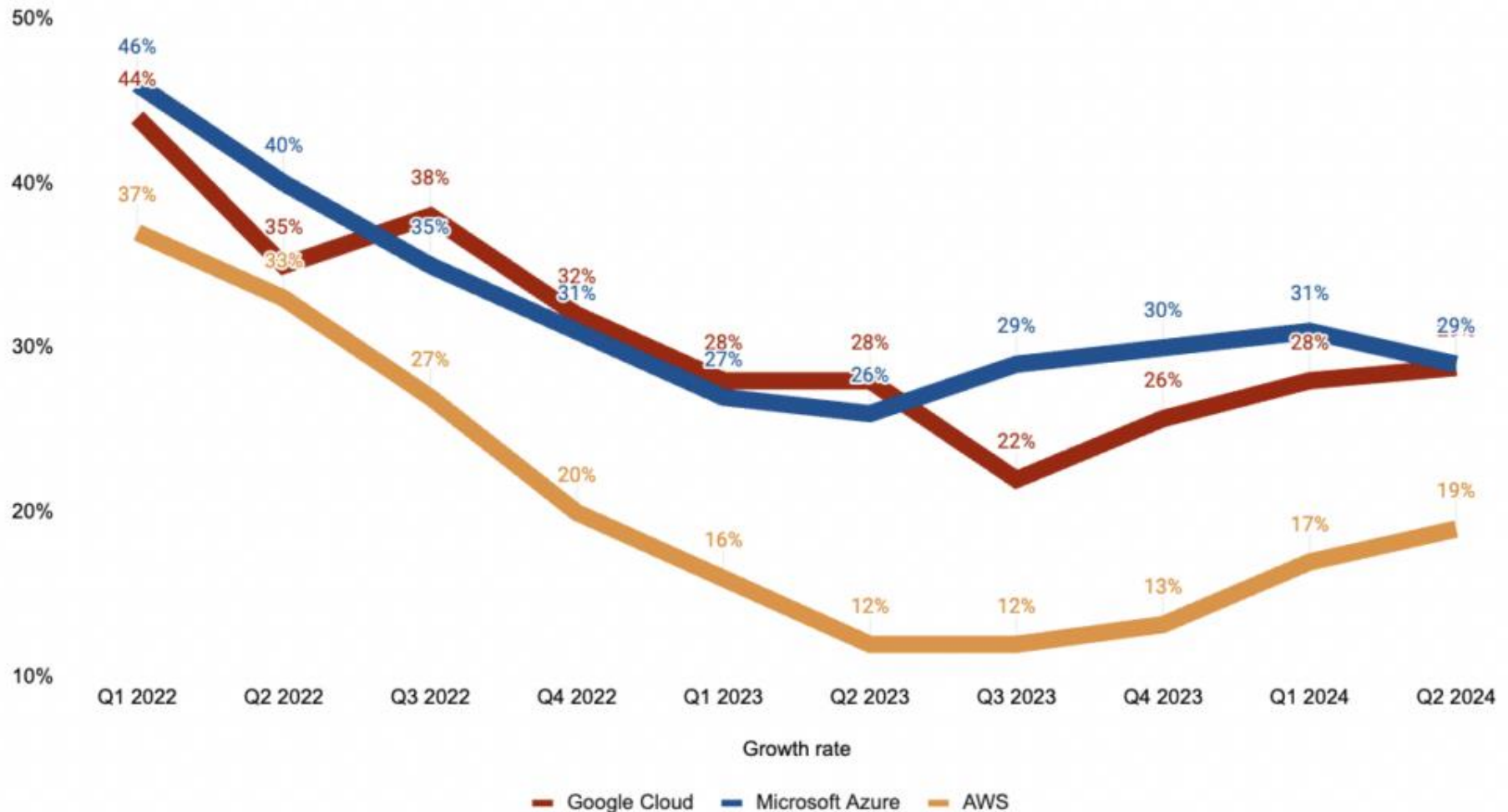
(IaaS, PaaS, Hosted Private Cloud)



Source: Synergy Research Group

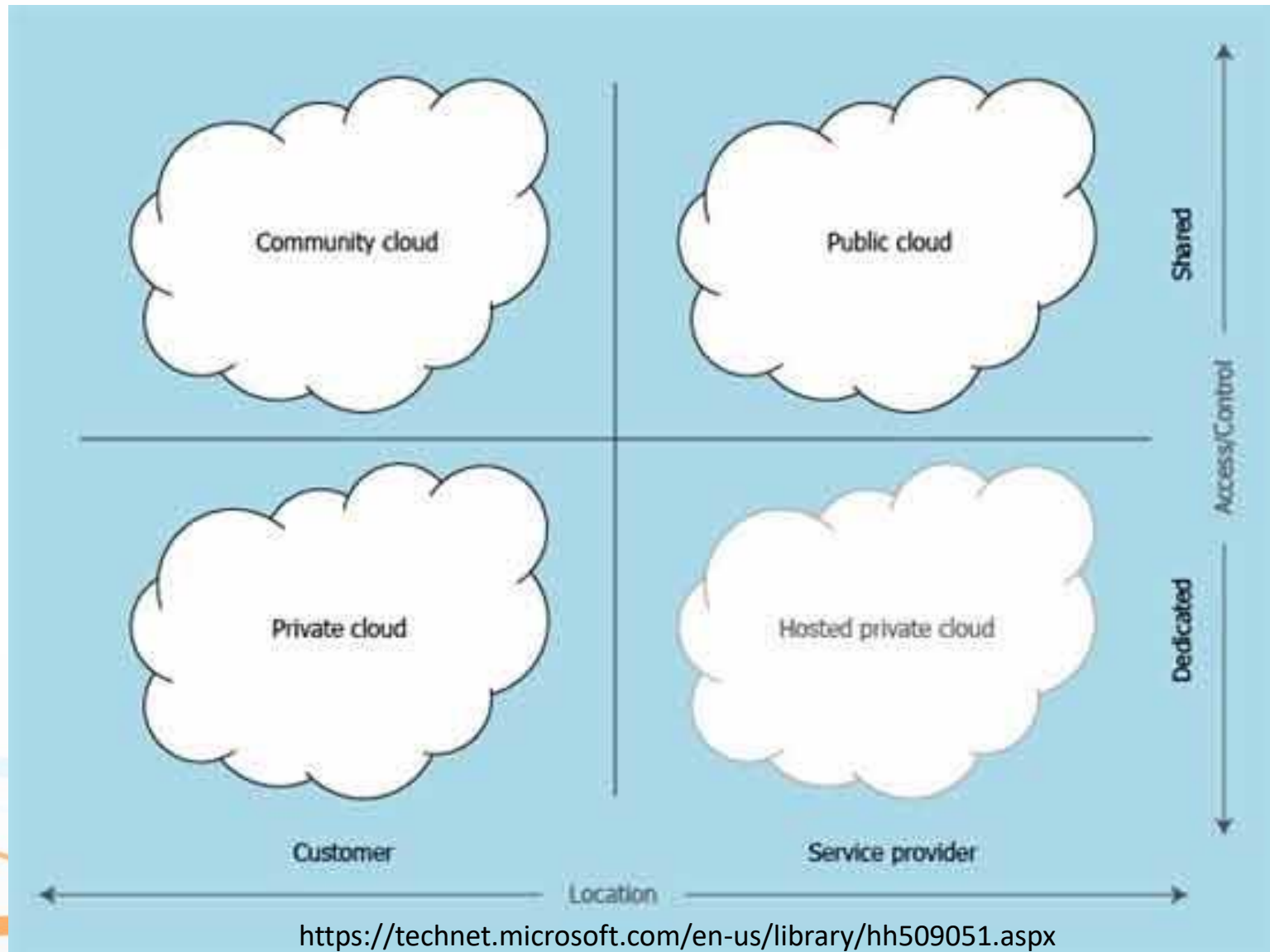
AWS, Azure, GCP growth fueled by AI

AWS, Azure and GCP growth rates 2022-2024

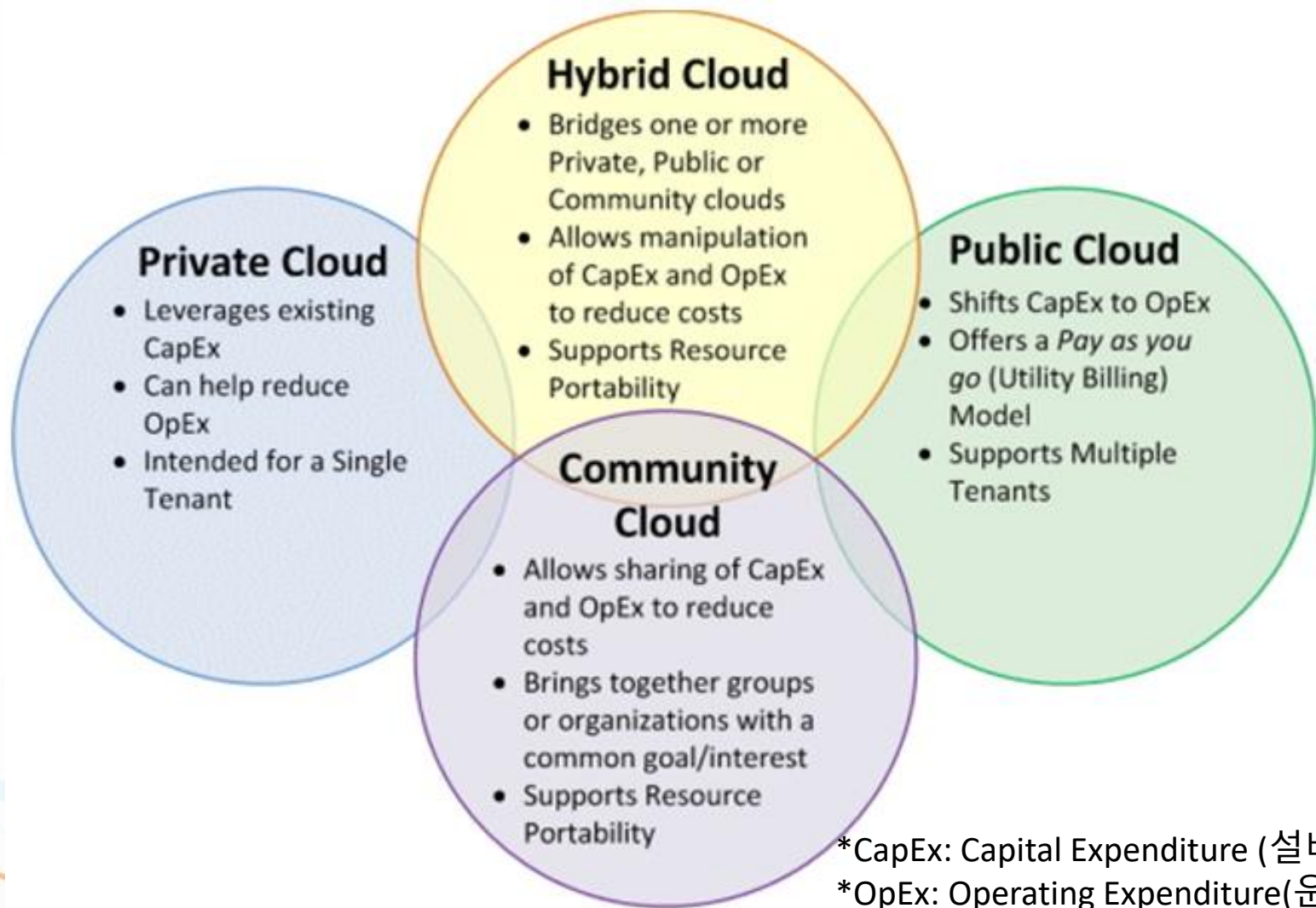


Cloud Computing Deployment Models

Cloud Computing Deployment Models



Cloud Computing Deployment Models



*CapEx: Capital Expenditure (설비투자비용)

*OpEx: Operating Expenditure(운영비용)

Cloud Computing Deployment Models

▶ Public cloud

- ❖ The **cloud infrastructure** is provisioned for open use by the **general public**
- ❖ It may be owned, managed, and operated **by a business, academic, or government organization, or some combination of them**

▶ Private cloud

- ❖ The **cloud infrastructure** is provisioned for exclusive use by **a single organization** comprising multiple consumers (e.g., business units)
- ❖ It may be owned, managed, and operated **by the organization, a third party, or some combination of them**, and it may exist on or off premises

Cloud Computing Deployment Models

▶ Community cloud

- ❖ The **cloud infrastructure** is provisioned for **exclusive use by a specific community** of consumers from organizations **that have shared concerns**

- e.g., mission, security requirements, policy, and compliance considerations

- ❖ It may be owned, managed, and operated **by one or more of the organizations** in the community, a third party, or some combination of them, and it may exist on or off premises

▶ Hybrid cloud

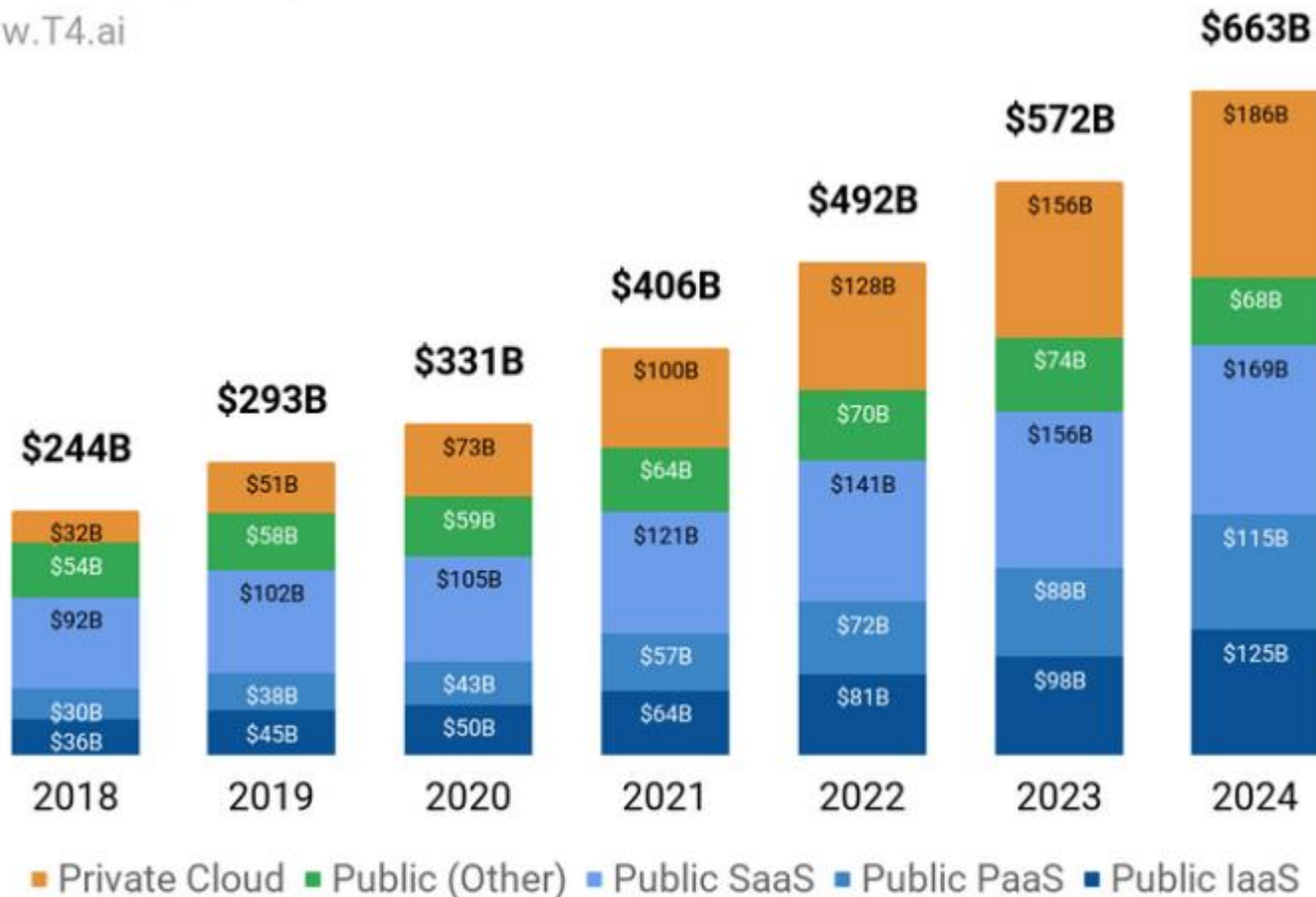
- ❖ The **cloud infrastructure** is a **composition of two or more distinct cloud infrastructures** (private, community, or public) that remain unique entities, but are bound together by standardized or proprietary technology that enables data and application portability

- e.g., cloud bursting for load balancing between clouds

Cloud Computing Market Share

Cloud Computing Market Size, 2018-2024

www.T4.ai



Desired Features of Cloud Service

▶ Self-Service

- ❖ Consumers of cloud computing services are able to access **resource on-demand, nearly instant access to resources**

▶ Multi-tenancy

- ❖ a single instance of software running on a server are shared by multiple tenants(groups of users)

▶ Per-Usage Metering and Billing

- ❖ **Services must be priced on a short term basis**
 - e.g., by the hour
- ❖ **Allowing users to release** (and not pay for) **resources as soon as they are not needed**

Desired Features of Cloud Service

▶ Elasticity

- ❖ Cloud computing gives the **illusion of infinite computing resources** available on demand
- ❖ **Additional resources can be**
 - **Provisioned**, possibly **automatically**, when an application load increases
 - Released when load decreases (**scale up and down**)

▶ Customization

- ❖ Allowing users to deploy **specialized virtual appliances** and to **be given privileged** (root) **access** to the virtual servers

▶ SLA

- ❖ Service robustness guarantee



Thank You!!



Any Questions ??